

7) ISA



↳ is like a menu: 'what?'

ADD : add two numbers

SUB : subtract two numbers.

LOAD : bring data from memory.

8) Microarchitecture → like a kitchen:

↳ It executes those ISA instructions.

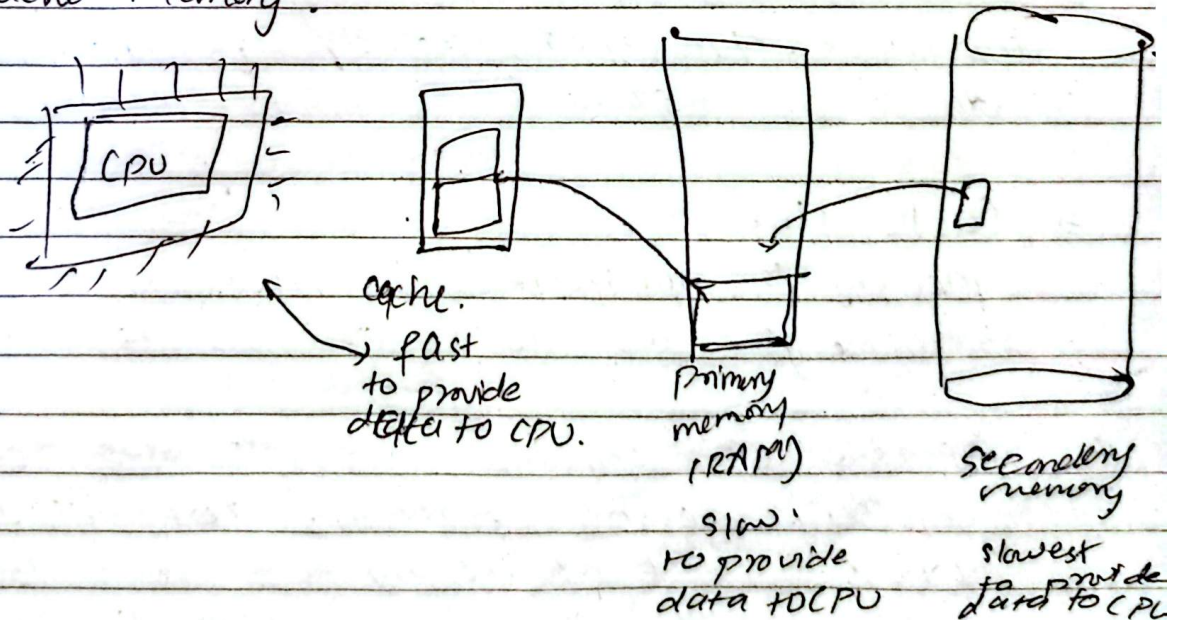
It handles how it is converted.

Day 2

Memory → Cache → Performance.

Main goal: why two program having same computation can have very different performance.

Cache Memory.



cost:	more costly	costly	cheaper.
size:	smallest	smaller	big
Access time:	fastest	fast	slow.

- CPU contacts cache memory for the ~~data~~ ^{data} required for it. If that data available in the cache memory, then this condition is cache hit.
- The time required to determine whether data is present there or not, is called hit latency.
- If the CPU required data is not available in cache memory, then the condition is cache miss. Then CPU contacts the main memory for that data, if available it takes it, if not then it need to contact the secondary memory.
- Then, the data's copies will be stored in primary memory and cache memory, so that if required again CPU can load it faster.

∴ The most frequently used / recently used data is stored in cache memory.

Cache line:

memory is transferred in blocks called lines.

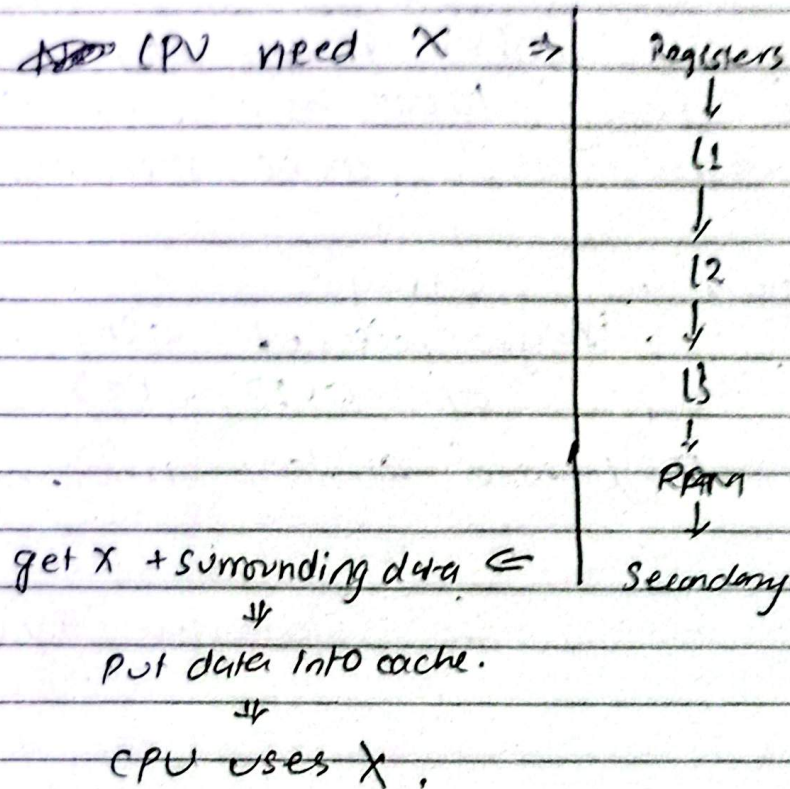
memory: 100 101 102 103 104 105 106

If the CPU requires 102, ~~it does~~ cache memory doesn't fetch only 102. It fetches whole block of memory from 100 - 106.

⑥ why bring extra data?

∵ Because program often access ~~near~~ ^{nearby} data.

L1, L2, L3 are architectural design choice.



2) Locality of Reference

Program CPU tends to access the same data or access data that are close together, this tendency is locality of reference.

Types:

- 1) Temporal Locality: Same things use again. Use recently, use ~~for~~ again soon, so it remains stored in the cache memory.

Like solving math problem.

Use calculator again & again, we don't put it inside the bag after every calculation. So, in computer it happens in loop:

```

for (i = 1; i < 100; i++) {
    sum = sum + i;
}
  
```

2. Spatial Locality.

If a program use one memory location, then there is higher probability that it will use nearby locations also., so load the block in the cache memory.

happen in Array:

{ if you accessed $a[0]$, high chance of $a[1]$, $a[2]$ may used in loop containing array.

- Cache memory is implemented by using SRAM.

Static-SRAM	Dynamic-DRAM
- Cache memory - very fast - costly.	- Main memory. - fast. - Cheaper than SRAM